

Reviewer Pack — Minimal Representation Complexity of Boolean Functions in Te...

Entry 7b889ecfe487 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Representation Complexity of Boolean Functions in Terms of Classical Groups

Entry ID: 7b889ecfe487 **Verdict:** INCONCLUSIVE **Recorded:** 2026-06-07 17:44:26 UTC

1. Conjecture

Field A (mathematical branch): Classical Group Theory (Representation Theory) **Field B** (complexity object): Boolean Functions (Circuit Complexity)

Statement:

The minimal representation complexity, measured by the dimension of the smallest irreducible representation, of a boolean function f is linearly correlated with its circuit size $s(f)$, such that $\dim(\text{Irr}(f)) = \Theta(s(f)^2)$.

Rationale (proposer's reasoning):

Classical group theory has a rich structure of representations that can be used to encode Boolean functions. The irreducibility and dimension of these representations could potentially capture the complexity of computing the function, providing a new perspective on circuit complexity.

Taxonomy category: c003b_cumulative_entropy/SC2#c49 (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 101c5eec4b9acc34

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The conjecture is supported if, for all boolean functions f with size $n \leq 40$, the correlation coefficient between $\dim(\text{Irr}(f))$ and $s(f)^2$ across 30 random seeds is above 0.7.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	UNCERTAIN	1.00	UNCERTAIN	HITS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	UNCERTAIN	1.00	UNCERTAIN	HITS
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: ADJACENT_OK against 3 hits across arXiv + Semantic Scholar + ECCC

Search queries (2): - "classical group theory" AND "boolean function complexity" - "irreducible representation" IN BOOLEAN GROUPS AND "boolean function", "{circuits size} in boolean functions AND classical groups"

Top relevant hits considered: - [http://arxiv.org/abs/2406.18700v4] Structure of sparse Boolean functions over Abelian groups, and its application to testing - [http://arxiv.org/abs/1911.04516v1] Boolean lattices in finite alternating and symmetric groups - [http://arxiv.org/abs/1303.1290v1] Symmetry groups of Boolean Functions

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution:** rc=124, elapsed=240.4s

5.1 Generated Python source

```
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_boolean_function(n):
        return [random.choice([0, 1]) for _ in range(2**n)]

    def circuit_size(f):
        # Simplified DPLL solver to estimate circuit size
        n = len(f)
        if n == 1:
            return 1
        subproblems = [f[:n//2], f[n//2:]]
        sizes = [circuit_size(sub) for sub in subproblems]
        return max(sizes) + 1

    def irreducible_representation_dimension(f):
        # Placeholder function to simulate computation of dimension
        n = len(f)
        if n == 1:
            return 1
        return n * (n - 1) // 2

    n_values = [5, 10, 15, 20, 30, 40]
    metric_values = []
    instances_tested = 0
    n_max = 0
```

```

for n in n_values:
    for _ in range(5): # Ensure at least 30 instances per seed
        f = generate_boolean_function(n)
        dim_irr = irreducible_representation_dimension(f)
        s_f = circuit_size(f)
        metric_values.append(dim_irr / (s_f ** 2))
        instances_tested += 1
        n_max = max(n_max, n)

if len(metric_values) < 30:
    return {
        "metric_name": "dim(Irr(f)) / s(f)^2",
        "metric_value": None,
        "instances_tested": instances_tested,
        "n_max": n_max,
        "conjecture_holds": False,
        "counterexample": "insufficient_instances"
    }

correlation_coefficient = sum((x - mean) * (y - mean) for x, y in zip(metric_values, metric_va
                             math.sqrt(sum((x - mean) ** 2 for x in metric_values) * sum((y - mea
mean_value = sum(metric_values) / len(metric_values)

return {
    "metric_name": "dim(Irr(f)) / s(f)^2",
    "metric_value": correlation_coefficient,
    "instances_tested": instances_tested,
    "n_max": n_max,
    "conjecture_holds": correlation_coefficient > 0.7,
    "counterexample": ""
}

if __name__ == "__main__":
    seeds = [int(x) for x in sys.argv[1:]] or [2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43,

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    mean_value = sum(r["metric_value"] for r in results if r["metric_value"] is not None) / len([r
    std_value = math.sqrt(sum((r["metric_value"] - mean_value) ** 2 for r in results if r["metric_
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if all(r["conjecture_holds"] for r in results):
        print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_fra
    elif any(not r["conjecture_holds"] for r in results):
        first_failing_seed = next(seed for seed, result in zip(seeds, results) if not result["con
        print(f"RESULT: FALSIFIED counterexample=\"\" first_failing_seed={first_failing_seed}")
    else:
        print(f"RESULT: INCONCLUSIVE reason=insufficient_data")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

TIMEOUT after 240s

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test timed out before producing data, which means it was unable to complete the required computations to verify the conjecture. | next: Re-run the test with increased time limits or optimize the code to ensure it completes within the given time frame.

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	13725
2	preregistration	ollama_remote	glm4:latest	0	0	10167
3	novelty	ollama_remote	glm4:latest	0	0	8232
4	novelty	ollama_remote	glm4:latest	0	0	9500
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	18272
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	10562
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	9593
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	11464
9	judge	ollama_remote	glm4:latest	0	0	25351

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 116866 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in *research/audit/7b889ecfe487.jsonl*)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in *research/audit/7b889ecfe487.jsonl* for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: *research/pvsnp_sandbox/test_*.py* (per-cycle)

Replay tarball: *research/replay/7b889ecfe487.tar.gz* (if generated)